

Programs on Strings:-

8 a. Write a C Program to read two strings and combine them without using string built-in functions.

```
#include<stdio.h>

void main(void)
{
    char str1[25], str2[25];
    int i=0, j=0;
    printf("\n Enter First String:");
    gets(str1);
    printf("\n Enter Second String:");
    gets(str2);
    while(str1[i]!='\0')
        i++;
    while(str2[j]!='\0')
    {
        str1[i]=str2[j];
        j++;
        i++;
    }
    str1[i]='\0';
    printf("\n Concatenated String is %s",str1);
}
```

Output:-

```
Enter First String: c programming lab
Enter Second String: UNIX and shell scripts
Concatenated String is c programming labUNIX and shell scripts
```

8 b. Write a C Program to read two strings and compare them without using string built-in functions.

```
#include <stdio.h>
```

```

int main ()
{
int count1 = 0, count2 = 0, i;
char string1[30], string2[30];
printf ("Enter the First string\n");
gets (string1);

printf ("Enter the Second string\n");
gets (string2);

while (string1[count1] != '\0')
    count1 ++;

while (string2[count2] != '\0')
    count2 ++;

i = 0;

while (string1[i] == string2[i] && string1[i] != '\0')
{
    i ++;
}

if (string1[i] > string2[i])
    printf ("First string is greater than Second string\n");
else if (string1[i] < string2[i])
    printf("Second string is greater than First string\n");
else
    printf ("Both strings are EQUAL\n");

return 0;
}

```

Output:-

```

Enter the First string
hello
Enter the Second string
hai
First string is greater than Second string

```

8.c. Write a C Program to demonstrate the use of built-in string functions like strcpy(),strcat(),strcmp() and strlen()

```
#include <stdio.h>
#include <string.h>
int main()
{
    char str1[100], str2[100];
    // Demonstrate strlen()

    printf("Enter a string: ");
    scanf("%s", str1);
    printf("Length of the string: %lu\n",

    strlen(str1)); // Demonstrate strcpy()

    strcpy(str2, str1);
    printf("Copied string using strcpy():

    %s\n", str2); // Demonstrate strcmp()

    printf("Enter another string: ");
    scanf("%s", str2);
    int cmpResult = strcmp(str1, str2);
    if (cmpResult == 0)
        printf("Strings are equal.\n");
    else if (cmpResult < 0)
        printf("First string is less than the second
        string.\n"); else
        printf("First string is greater than the second

        string.\n"); // Demonstrate strcat()

    strcat(str1, str2);
    printf("Concatenated string using strcat():

    %s\n", str1); return 0;
}
```

Output:-

Enter a string: hello
Length of the string: 5
Copied string using strcpy(): hello
Enter another string: hai
First string is greater than the second
string. Concatenated string using
strcat(): hellohai